Opponent-Adjusted Evaluation of Pass Blocking and Pass Rushing Performance in the NFL

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Abstract

Quantifying pass protection and pass rushing at the player level remains a core challenge in football analytics. Box-score outcomes such as sacks and hits contain useful signal, but they are sparse, context-dependent, and typically unadjusted for opponent strength. Using 2021 regular-season Hudl tracking data, we construct a blocker–rusher interaction dataset and estimate two ridge-regularized Bradley–Terry (BT) models: (i) a binary win/loss model using a 2.5-second pass-rush win definition, and (ii) a multinomial severity model over `loss`/`win`/`hit`/`sack`. The modeling table contains 153,138 interactions across 33,283 pass plays in 266 games.

On an ordered 80/20 holdout split (n_{test} = 30,628), BT improves win-model log loss versus a global baseline (+0.00682) and a matchup baseline (+0.00141). For severity, the primary endpoint is multiclass log loss, where BT improves over both a global class-frequency baseline (+0.00755) and an OL/DL matchup baseline (+0.00137). End-to-end game-level bootstrap intervals support positive improvements on key endpoints. As an external construct check, we compare BT rankings with 2021 AP All-Pro selections using rank AUC and enrichment@K (K equals the number of positives in each role/accolade slice), benchmarked against a raw empirical win-rate ranking baseline. Severity BT shows strong alignment on first+second-team labels (AUC = 0.866 for blockers; 0.752 for rushers; enrichment@K = 6.96× and 12.65×, respectively), while the raw win-rate baseline yields enrichment@$K = 0$ in these slices.

1 Introduction

Pass protection and pass rushing are central to modern NFL efficiency and roster construction, but player-level evaluation in the trenches remains difficult. For rushers, box-score outcomes such as sacks and hits are rare and heavily mediated by context, including quarterback time-to-throw, coverage, and game situation. For blockers, the inverse problem holds: a lineman can execute consistently without generating a direct box-score statistic. Aggregate summaries often blur individual ability with opponent quality, help structure, or team environment.

To better isolate line play, ESPN introduced pass block win rate (PBWR), which measures whether a blocker sustains a block for 2.5 seconds on a pass play (Burke, 2018). More recent tracking-based work has moved beyond sacks alone by modeling how pressure develops in space and time. For example, STRAIN summarizes how quickly defenders close space to the quarterback over time (Nguyen et al., 2023). These approaches provide richer information than sacks alone, but they are usually reported as season-level summaries and do not directly deliver opponent-adjusted player ratings for blockers and rushers in one common framework.

Our goal is to evaluate offensive and defensive line play within one opponent-adjusted paired-comparison framework. We treat each blocker–rusher engagement as a matchup between a pass protector and a pass rusher, then fit regularized Bradley–Terry models (Bradley and Terry, 1952; Friedman et al., 2010; Glickman and Jones, 2025). We estimate two separate ridge-regularized BT models: one for the binary 2.5-second win/loss task and one for an outcome-weighted severity task. This separation lets us validate each task against its own baseline and quantify uncertainty in a way that matches the estimand.

Contributions. Our main contributions are:

1. A paired-comparison framework that evaluates blockers and rushers jointly while preserving role-specific interpretation.
2. Separate regularized BT models for binary win/loss and multi-outcome severity.
3. Chronological out-of-sample validation against task-specific baselines, including bootstrap uncertainty for predictive performance.
4. External rank-based validation against AP All-Pro selections using AUC and enrichment@K, benchmarked against a raw win-rate baseline.
5. End-of-season leaderboards and cumulative weekly path uncertainty for longitudinal interpretation.

2 Data

We use NFL player-tracking data from the 2021 regular season provided by Hudl. Tracking coordinates are recorded at 10 Hz and are accompanied by event annotations marking events such as blocking engagements, pass attempts, and sacks.

We restrict attention to dropbacks by retaining plays containing either a forward pass event or a quarterback sack event. Within each retained play, we keep offensive linemen, identified pass rushers, and the quarterback. For each frame t, we compute the Euclidean distance between the quarterback and any non-quarterback player:

$$d_{p,t} = \sqrt{(x_{p,t} - x_{\mathrm{QB},t})^2 + (y_{p,t} - y_{\mathrm{QB},t})^2}. \quad (1)$$

Our unit of analysis is a blocker–rusher interaction, defined from the engagement labels in the tracking data. We generate a binary double-team indicator and set it to 1 when multiple blockers are assigned to the same rusher within overlapping time windows.

After filtering and labeling, the modeling table contains 153,138 blocker–rusher interactions across 33,283 pass plays in 266 games, with 620 rushers and 348 blockers. The double-team rate is 42.70%.

3 Outcomes

We model two targets:

1. **Win/Loss target** (`win_target`): indicator of whether the rusher wins within the 2.5-second definition.
2. **Severity target** (`severity_outcome`): multinomial outcome in $\{\texttt{loss}, \texttt{win}, \texttt{hit}, \texttt{sack}\}$, with scalar severity mapping $0, 0.2, 0.4, 1.0$ for weighted evaluation.

We assign one outcome per interaction using severity priority `sack` > `hit` > `win` > `loss`. A `win` occurs if the rusher gains a decisive advantage within 2.5 seconds, operationalized as becoming closer to the quarterback than the blocker before the 2.5-second threshold. A `sack` is taken directly from event annotations. A `hit` is recorded when the quarterback is contacted behind the line of scrimmage before pass release without a sack. If none of these events occurs, the interaction is labeled `loss` from the rusher's perspective.

Observed frequencies in the full table are:

$$\mathbb{P}(\texttt{loss}) = 0.732, \quad \mathbb{P}(\texttt{win}) = 0.253, \quad \mathbb{P}(\texttt{hit}) = 0.0091, \quad \mathbb{P}(\texttt{sack}) = 0.0063.$$

4 Modeling Framework

4.1 Win/Loss Ridge BT

For interaction t, with rusher $i(t)$, blocker $j(t)$, and double-team indicator D_t, the binary BT model is

$$\operatorname{logit} \mathbb{P}(Y_t = 1) = r_{i(t)} - b_{j(t)} + \delta D_t,$$

where $Y_t = 1$ indicates rusher win.

We estimate parameters via ridge-penalized logistic regression:

$$\arg\min_{\theta} \left\{ -\ell(\theta) + \lambda \|\theta\|_2^2 \right\},$$

with λ selected by cross-validation on a log-spaced grid.

Throughout the paper, we report raw BT scores without affine rescaling so that figures and tables remain directly tied to fitted model coefficients.

4.2 Severity Ridge BT

For severity classes $c \in \{\texttt{loss}, \texttt{win}, \texttt{hit}, \texttt{sack}\}$, we fit a ridge-regularized multinomial BT model:

$$\mathbb{P}(C_t = c) = \frac{\exp(\eta_{t,c})}{\sum_{c'} \exp(\eta_{t,c'})}, \qquad \eta_{t,c} = r_{i(t),c} - b_{j(t),c} + \delta_c D_t.$$

The scalar severity prediction is the expected score under class probabilities, using the mapping `loss` $= 0$, `win` $= 0.2$, `hit` $= 0.4$, and `sack` $= 1.0$.

4.3 Validation Baselines

We sort interactions by `game_id`, `play_id`, and `event_game_index`, then apply a deterministic 80/20 ordered split (train = 122,510, test = 30,628).

Win model baselines:

1. **Global mean baseline**: train-set rusher win proportion.
2. **Matchup baseline (logit mean)**: smoothed rusher/blocker components combined on the logit scale with prior strength 25.

Severity model baselines:

1. **Global scalar baseline**: train-set expected severity score.
2. **Global multiclass baseline**: train-set marginal class probabilities.
3. **Matchup multiclass baseline**: smoothed rusher and blocker class profiles (prior strength 50) combined with a multinomial logit-mean; scalar matchup expectations are derived from the resulting class probabilities.

4.4 Uncertainty

We report two complementary uncertainty procedures (Efron and Tibshirani, 1994):

1. **End-to-end bootstrap** ($B = 400$): resample games, refit with fixed λ selected from full-data CV, and recompute validation metrics and player ratings.
2. **Weekly path bootstrap** ($B = 100$): for each cumulative week checkpoint, resample past games and refit to obtain path-level uncertainty bands.

5 Results

5.1 Model Fit and Validation

Cross-validated ridge selected $\lambda_{\min} = 1.31 \times 10^{-4}$ for win BT and $\lambda_{\min} = 1.04 \times 10^{-4}$ for severity BT. The win BT fit retains 970 nonzero terms at $\lambda_{\min}$.

Table 1 summarizes holdout performance and bootstrap uncertainty. For severity, multiclass log loss is the primary endpoint: BT improves versus both global and matchup baselines.

5.2 Ratings and Leaderboards

Figure 1 shows raw BT score distributions by role for both tasks. Figure 2 reports top players by model and role (minimum 200 interactions to reduce small-sample volatility). Scores are

Table 1: Out-of-sample validation metrics for BT models and baselines. Improvements are baseline minus model (higher is better). For rows with intervals, 95% bounds are from the game-level end-to-end bootstrap distribution of improvement.

Task	**Metric**	**Model**	**Baseline**	**Improvement**	**95% bootstrap CI**
Win vs global	Brier	0.18547	0.18806	0.00259	[0.00225, 0.00446]
Win vs global	Log loss	0.55680	0.56362	0.00682	[0.00579, 0.01188]
Win vs matchup	Brier	0.18547	0.18598	0.00051	[0.00011, 0.00100]
Win vs matchup	Log loss	0.55680	0.55821	0.00141	[0.00025, 0.00258]
Severity (scalar)	MSE	0.01345	0.01354	0.00009	[0.00011, 0.00023]
Severity (multiclass, global)	Log loss	0.62717	0.63473	0.00755	[0.00834, 0.01559]
Severity (multiclass, matchup)	Log loss	0.62717	0.62854	0.00137	not estimated

interpreted within each model (win vs severity) rather than across models, because the two targets induce different scales. The win/loss and severity leaderboards are directionally similar for elite rushers but show clear differences for blockers, which is expected because the severity model upweights high-impact outcomes.

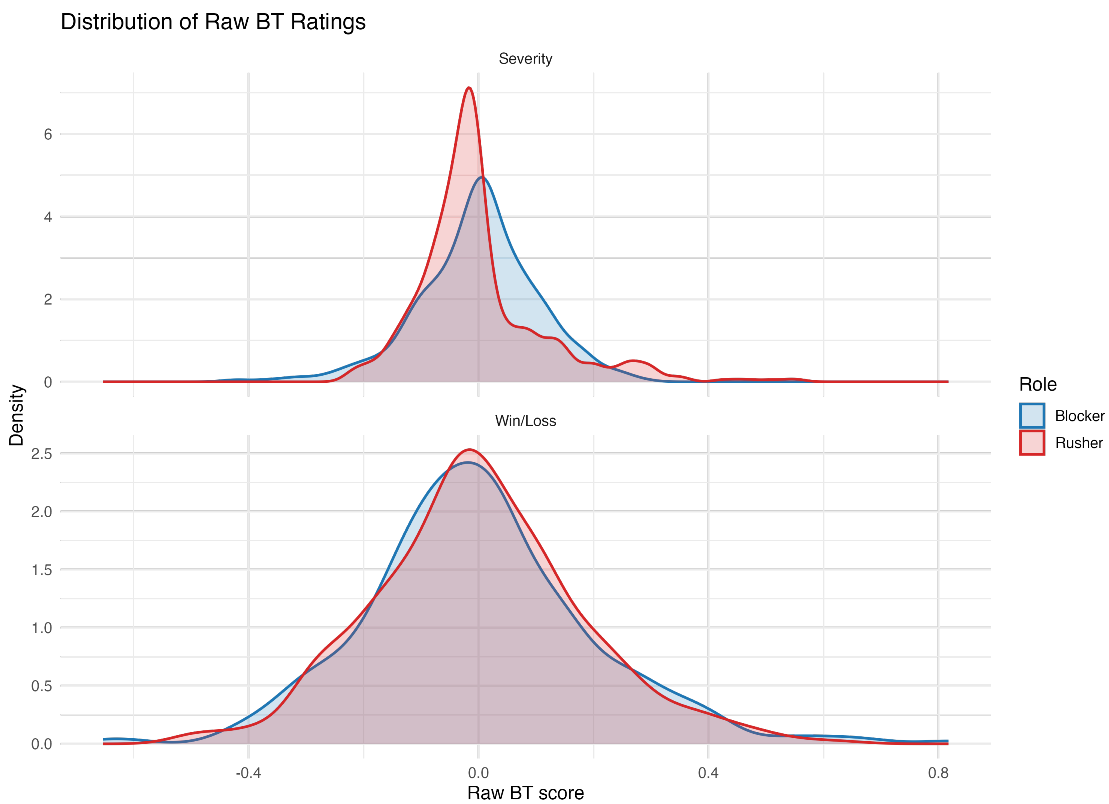

Figure 1: Distribution of raw BT scores by model and role.

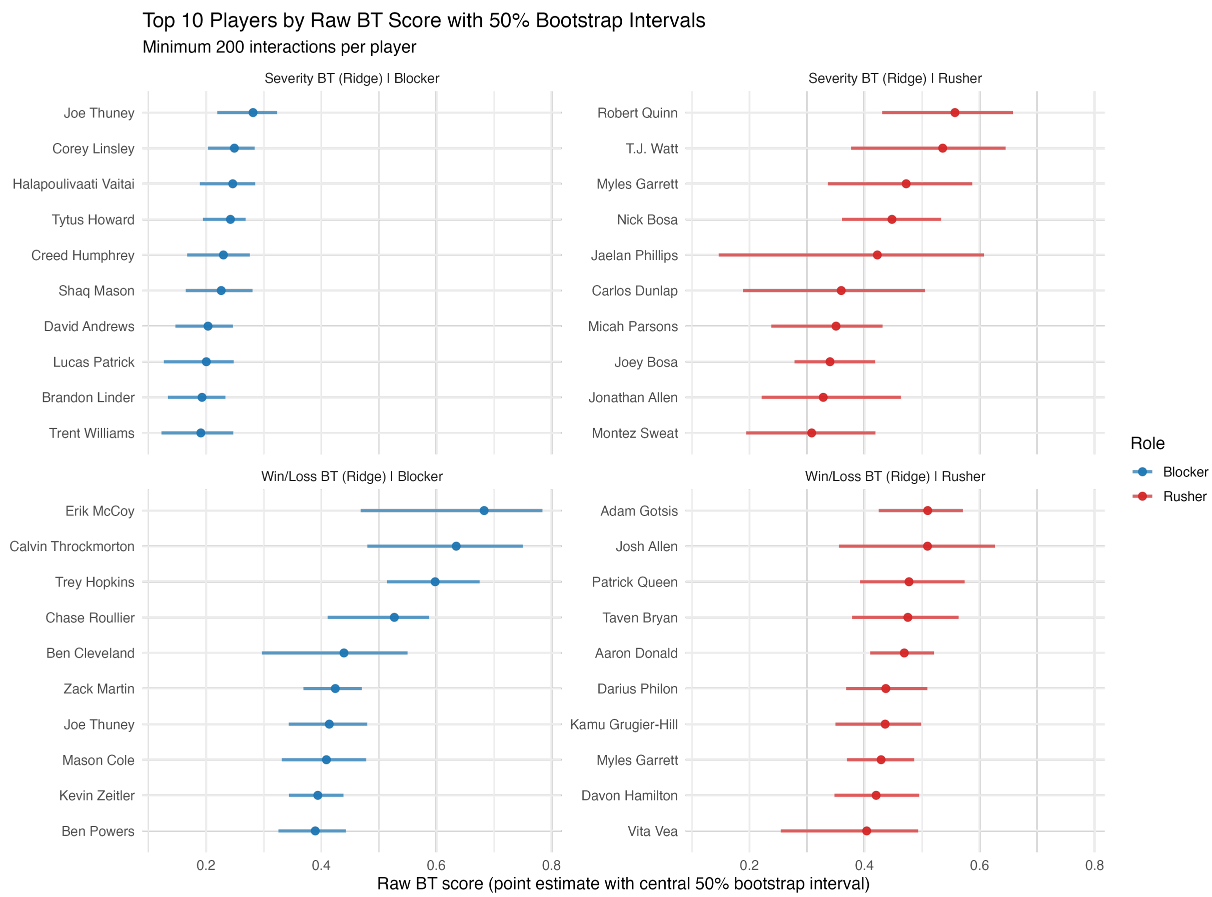

Figure 2: Top 10 players by raw BT score in each model-role panel (minimum 200 interactions). Points are estimates and horizontal bands are central 50% bootstrap intervals (25th–75th percentiles).

Table 2: Top five players by model and role (minimum 200 interactions). Values are raw BT scores.

Model	**Role**	**Player**	**Rating**
Win/Loss BT	Rusher	Adam Gotsis	0.510
Win/Loss BT	Rusher	Josh Allen	0.509
Win/Loss BT	Rusher	Patrick Queen	0.477
Win/Loss BT	Rusher	Taven Bryan	0.475
Win/Loss BT	Rusher	Aaron Donald	0.469
Win/Loss BT	Blocker	Erik McCoy	0.683
Win/Loss BT	Blocker	Calvin Throckmorton	0.635
Win/Loss BT	Blocker	Trey Hopkins	0.598
Win/Loss BT	Blocker	Chase Roullier	0.527
Win/Loss BT	Blocker	Ben Cleveland	0.440
Severity BT	Rusher	Robert Quinn	0.557
Severity BT	Rusher	T.J. Watt	0.536
Severity BT	Rusher	Myles Garrett	0.472
Severity BT	Rusher	Nick Bosa	0.448
Severity BT	Rusher	Jaelan Phillips	0.422
Severity BT	Blocker	Joe Thuney	0.282
Severity BT	Blocker	Corey Linsley	0.249
Severity BT	Blocker	Halapoulivaati Vaitai	0.246
Severity BT	Blocker	Tytus Howard	0.242
Severity BT	Blocker	Creed Humphrey	0.230

5.3 External Validation Against All-Pro Selections

To test whether BT rankings align with high-end expert accolade outcomes, we evaluate against 2021 AP All-Pro labels (first team and first+second team). We compare BT model rankings against a simple baseline ranking based on empirical raw win rates (rusher: $\mathbb{E}[\texttt{win\_target}]$, blocker: $\mathbb{E}[1 - \texttt{win\_target}]$). We report:

1. **Rank AUC**: probability that a randomly chosen All-Pro player is ranked above a randomly chosen non-All-Pro player. With labels $y_i \in \{0, 1\}$, scores s_i, $n_+ = \sum_i \mathbf{1}\{y_i = 1\}$, and $n_- = \sum_i \mathbf{1}\{y_i = 0\}$, we use the Mann–Whitney form

$$\mathrm{AUC} = \frac{1}{n_+ n_-} \sum_{i:y_i=1} \sum_{j:y_j=0} \Big[\mathbf{1}\{s_i > s_j\} + \tfrac{1}{2}\mathbf{1}\{s_i = s_j\}\Big].$$

2. **Enrichment@K**: precision@K divided by base rate, where K is the number of All-Pro positives in that role/accolade slice (equivalently, the AP selection-slot count for that slice):

$$\text{Enrichment@}K = \frac{\text{precision@}K}{n_+/n}, \qquad \text{precision@}K = \frac{\text{hits@}K}{K}.$$

   Here n is the number of players in the role/accolade slice.

Table 3 shows severity BT leads on AUC in three of four role/accolade slices and delivers the strongest enrichment@K for blockers (first+second team) and rushers (both accolade sets), indicating concentration of All-Pro players near the top of BT rankings relative to role-level prevalence. The raw win-rate baseline has moderate AUC in some slices but yields enrichment@$K = 0$ throughout, indicating no All-Pro selections captured in the corresponding top-K sets.

Table 3: All-Pro alignment metrics by model, role, and accolade set (2021 AP labels). K equals the number of positives in each role/set (the corresponding AP slot count).

Model	**Role**	**Accolade set**	K	**AUC**	**Enrichment@K**
Raw win-rate baseline	Blocker	First+Second Team	10	0.653	0.00
Raw win-rate baseline	Blocker	First Team	5	0.712	0.00
Raw win-rate baseline	Rusher	First+Second Team	14	0.718	0.00
Raw win-rate baseline	Rusher	First Team	7	0.702	0.00
Severity BT	Blocker	First+Second Team	10	0.866	6.96
Severity BT	Blocker	First Team	5	0.871	0.00
Severity BT	Rusher	First+Second Team	14	0.752	12.65
Severity BT	Rusher	First Team	7	0.816	25.31
Win/Loss BT	Blocker	First+Second Team	10	0.654	3.48
Win/Loss BT	Blocker	First Team	5	0.735	0.00
Win/Loss BT	Rusher	First+Second Team	14	0.768	6.33
Win/Loss BT	Rusher	First Team	7	0.749	0.00

5.4 Weekly Path Uncertainty

The weekly path bootstrap adds longitudinal uncertainty to the end-of-season summary. The regular-season dataset yields 18 cumulative weekly checkpoints for each player-model-role series. Figure 3 shows representative paths for high-impact rushers and blockers in each task.

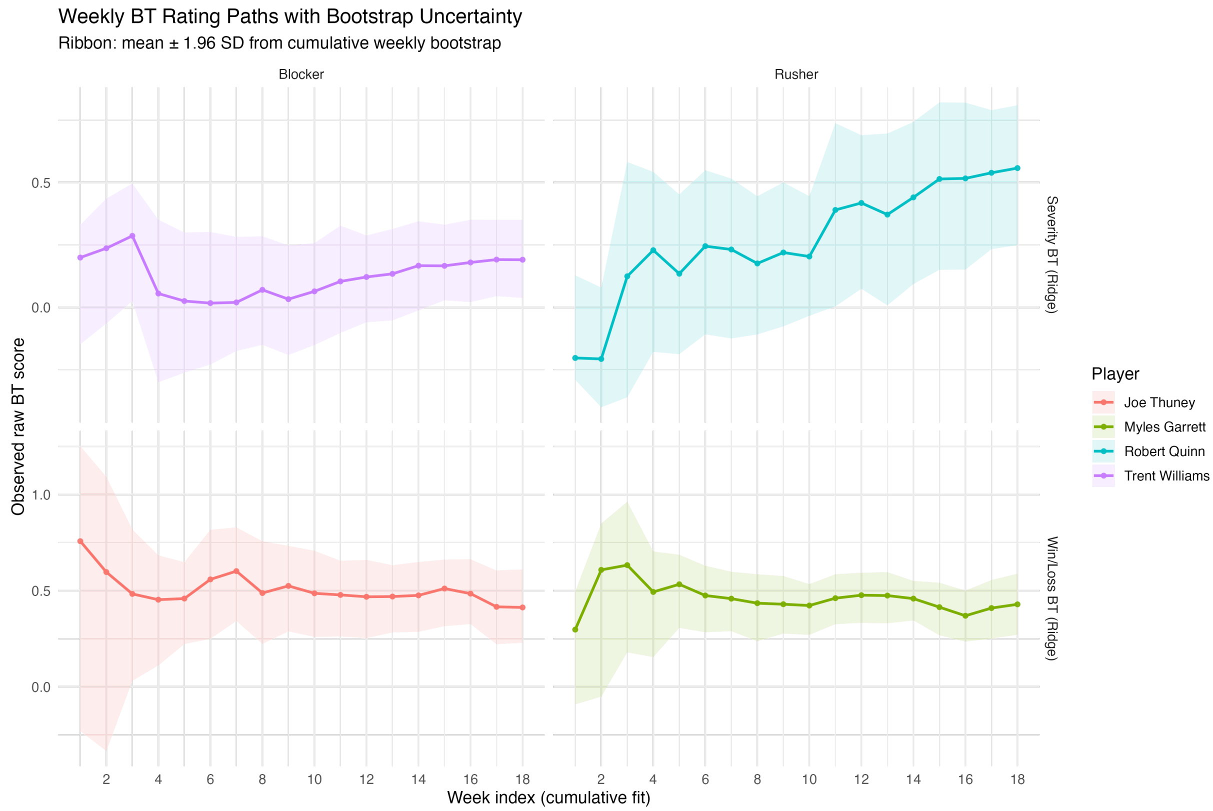

Figure 3: Weekly cumulative raw BT score paths with bootstrap uncertainty ribbons (mean $\pm$ 1.96 SD).

6 Discussion

The BT framework provides a unified, opponent-adjusted view of pass-rush and pass-protection performance with role-specific interpretation. Across holdout evaluations, both BT tasks improve on global baselines. On the primary severity endpoint (multiclass log loss), BT also improves on a stronger OL/DL matchup baseline. These results support regularized BT as an effective modeling family for both binary win/loss and severity-weighted evaluation.

6.1 Limitations and Future Work

Several limitations suggest clear extensions. First, the `win` label is based on a distance-to-quarterback rule and may not fully capture functional pressure quality. More realistic labels could incorporate pocket geometry, rush lane, and quarterback decision constraints.

Second, help mechanisms (chip blocks, tight-end/running-back assistance, and protection-slide structure) are only partially captured by a coarse double-team indicator. Third, sacks and pressure proxies are influenced by quarterback time-to-throw and play design, which are only indirectly represented. Finally, although BT adjusts for opponent strength, it does not explicitly model teammate effects or position-family hierarchy; hierarchical shrinkage and multi-season pooling are natural next steps.

6.2 Conclusion

Regularized BT models provide a stable and interpretable framework for NFL pass-rush and pass-protection evaluation. In this 2021 regular-season study, BT outperforms baseline approaches on both binary win/loss and outcome-severity tasks, with uncertainty quantified at both endpoint and weekly-path levels.

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